

UNIT 3

PROGRAMMING

FUNDAMENTALS

9TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

Basic Terminologies

HTML

Tags in HTML

Tags used in text

Cascading Style Sheets (CSS)

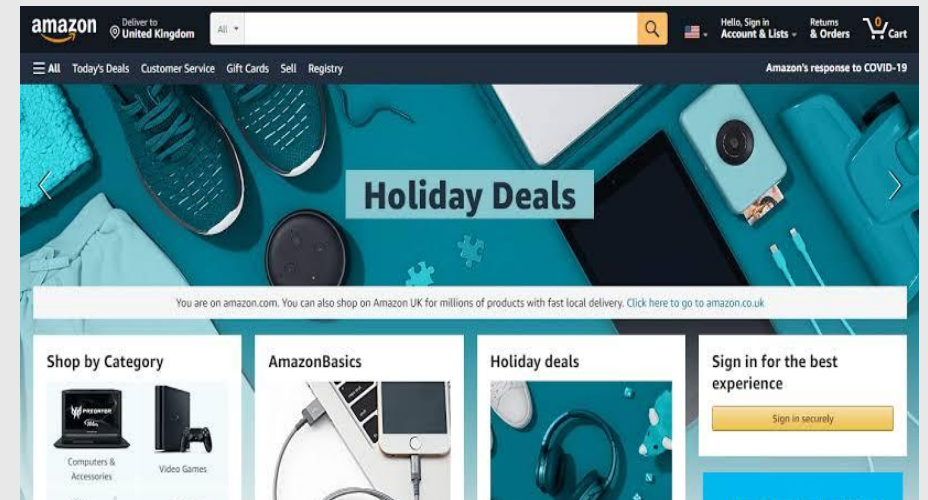
JavaScript (JS)

BASIC TERMINOLOGIES

Webpage – A single document or page on the internet. E.g. The "Contact Us" page on a website.



Website – A collection of many webpages linked together under one domain name. E.g. The entire Amazon site with all its pages.



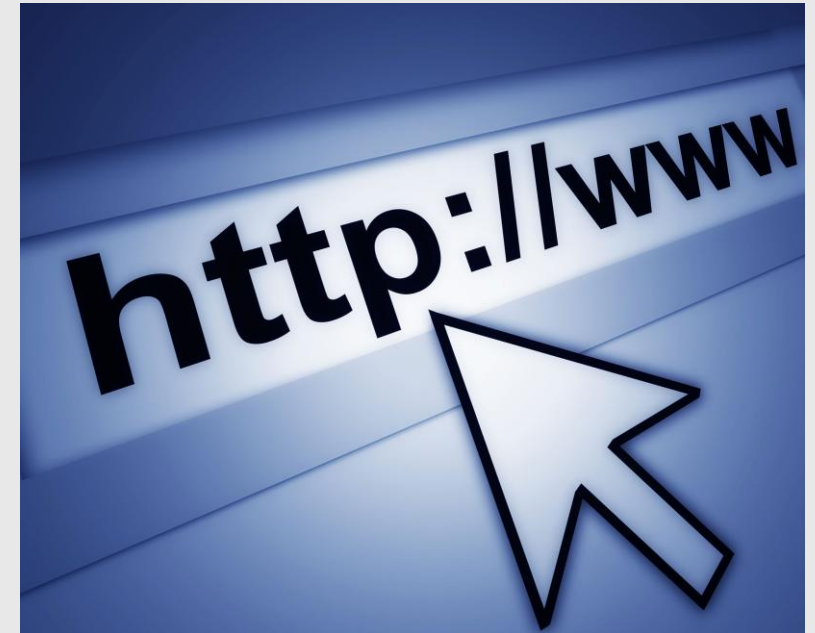
BASIC TERMINOLOGIES CONT....

URL: It is the web address you type in a browser to visit a specific webpage e.g. <https://www.google.com>

Breakdown of a simple URL:

<https://www.example.com/home>

- <https://> – Protocol (rules for transferring data)
- [www.example.com](https://www.example.com/home) – Domain name (website's address)
- [/home](https://www.example.com/home) – Path (specific page or file)

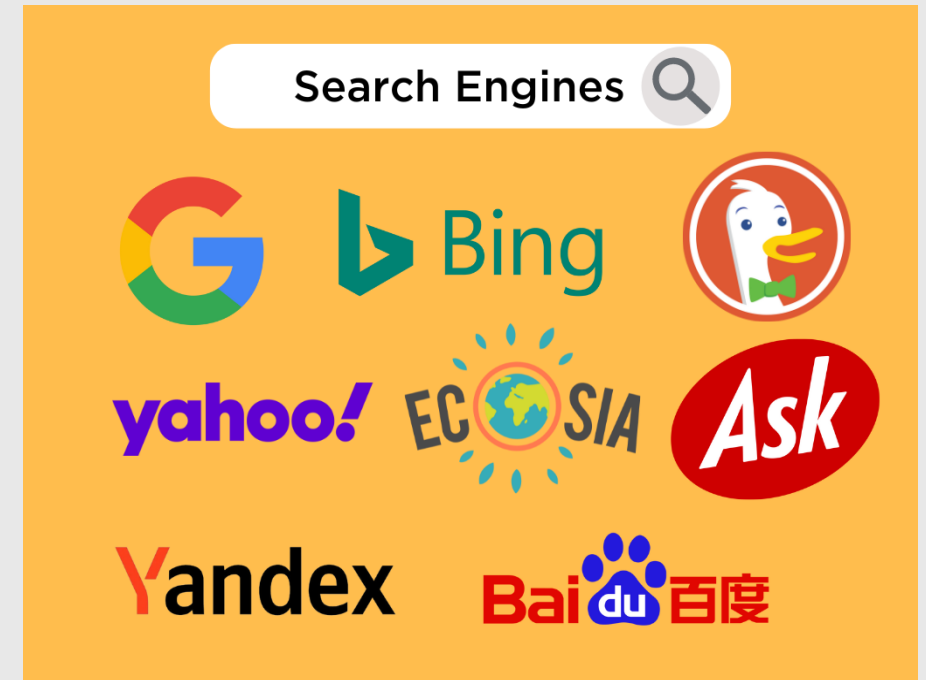


BASIC TERMINOLOGIES CONT....

Search Engine: It is a tool/website that helps you find information on the internet.

You type keywords or questions, and it shows you a list of relevant webpages.

Examples: Google, Bing, Yahoo, DuckDuckGo.



BASIC TERMINOLOGIES CONT....

Web Hosting Service: It is a service to stores your website's files (text, images, videos) on a server (a powerful computer) so that people can access it on the internet.

Without web hosting, your website cannot be viewed online.

Common web hosting providers: Hostinger, Bluehost, GoDaddy.



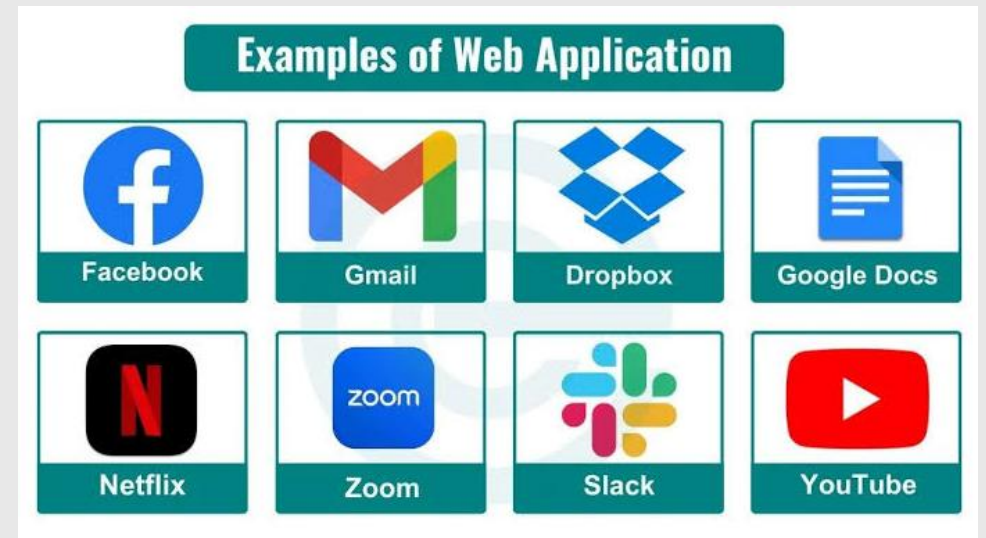
BASIC TERMINOLOGIES CONT....

Web Application: It is an interactive program that runs in browser and allows to do things (not just read information), like:

- *Send emails (Gmail)*
- *Edit documents (Google Docs)*
- *Shop online (Amazon)*
- *Book tickets (MakeMyTrip)*

Key difference from a normal webpage:

- **Webpage** = mostly read-only information (like a news article).
- **Web application** = lets you interact, enter data, and get results based on your actions.



BASIC TERMINOLOGIES CONT....

Static Website: A website where content stays the same for every user unless the developer manually changes the code.

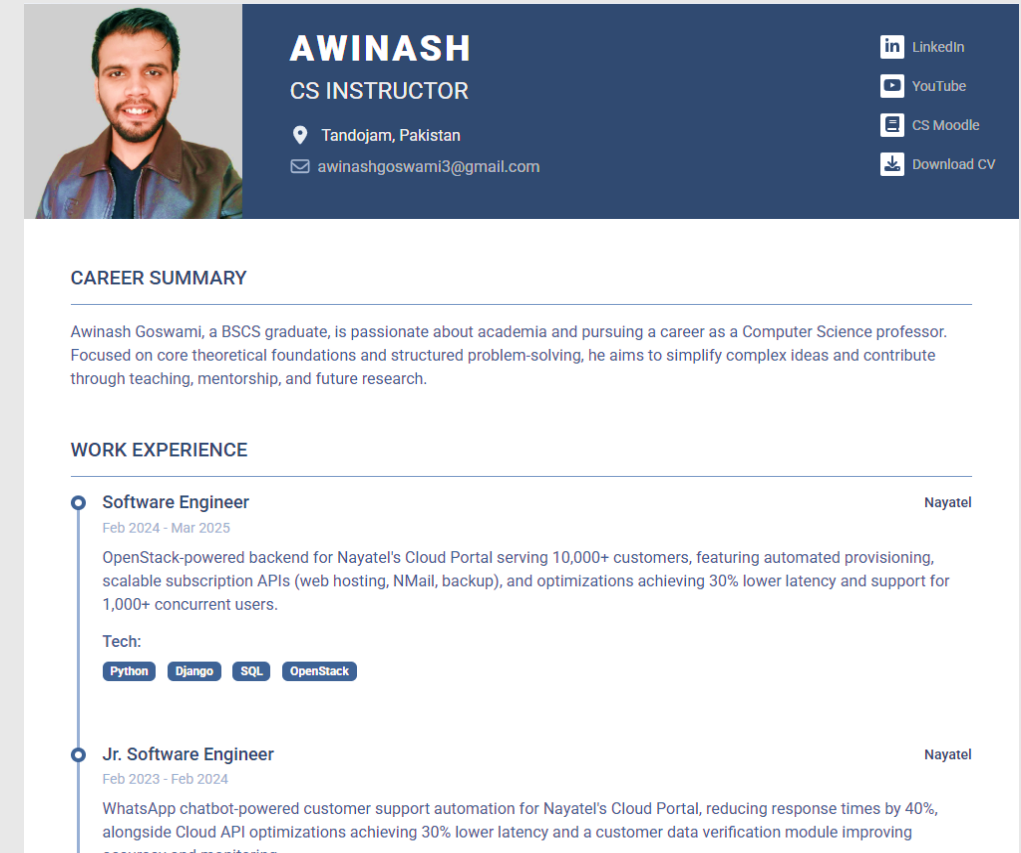
It is made using only HTML, CSS, and sometimes JavaScript (no databases or server-side processing).

Examples:

A personal portfolio page

A small business brochure site

A landing page with fixed information



BASIC TERMINOLOGIES CONT....

Dynamic Website: A website where content changes based on the user, time, or actions.

It uses server-side languages (like PHP, Python, or JavaScript) and databases to generate pages on the fly.

Content is not fixed – it is created in real-time when someone visits.

Examples:

- *Facebook (feed changes for each user)*
- *YouTube (recommendations based on your history)*
- *Amazon (product suggestions and personalized prices)*
- *Gmail (your inbox updates with new emails)*



BASIC TERMINOLOGIES CONT....

Front-End Development: It is the part of web development that deals with everything the user sees and interacts with directly in their browser. It focuses on the look, feel, and behavior of a website or web application.

Key Technologies:

- *HTML – Structure (headings, paragraphs, images)*
- *CSS – Styling (colors, fonts, layout)*
- *JavaScript – Interactivity (buttons, animations, forms)*

Examples of what front-end developers do:

- *Design menus, buttons, and forms*
- *Make a website look good on mobile and desktop*
- *Create smooth animations and pop-ups*
- *Ensure fast loading and easy navigation*



BASIC TERMINOLOGIES CONT....

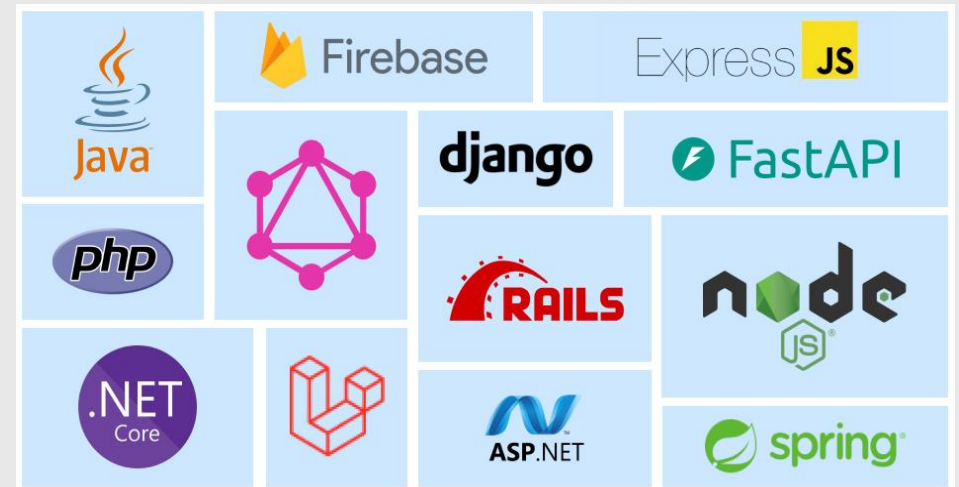
Back-End Development: It is the part of web development that deals with everything behind the scenes – the server, database, and application logic that users do not see. It makes sure the website works properly, stores data, and processes user requests.

Key Technologies:

- *Languages: Python, PHP, Java, Ruby, C#, JavaScript (Node.js)*
- *Databases: MySQL, PostgreSQL, MongoDB (to store user data)*
- *Servers: Apache, Nginx (to host and run the website)*

Examples of what back-end developers do:

- *Store user login details in a database*
- *Process online payments securely*
- *Send emails or notifications*
- *Fetch and send data to the front-end when a user clicks a button*



HTML

- Hypertext Markup Language (HTML) is used to define contents with help of tags for webpage.
- Tags are written with angle brackets '<' and '>'. For instance; <p> </p> tag used for paragraph and <h1> </h1> to define largest heading.
- The content is written between opening and closing tag such as <p> and </p> respectively.
- There are also some tag which don't need ending tags like
 <hr> for break line and horizontal line respectively.
- Actual Code example of tag and its content: <p> This is my first website </p> this line will display the in browser.
- You can write your html code in a notepad and run it in a browser.
- Other better software for the coding that help in minimizing your coding efforts are Visual Studio, NetBeans, etc.

TAGS IN HTML

- **html:** This is first tag of the HTML document, everything we see on browser lies inside the tag.
- **head:** This tag comes after <html> tag which is used to define metadata which is not for browser content but information about website like title and links like style.css and script.js etc.
- **title:** <title> tag is used define name of the website which we see on tab in a browser.
- **body:** <body> tag is one where we write our actual content like headings, paragraphs, links, buttons etc.
- **Tags for Headings:** There are six headings in html document from h1 to h6 as largest to smallest <h1> </h1>
- **Tag for Line Break:** To split a sentence into multiple lines a tag is used.
- **span:** tag is used to write content but it takes space of only content.
- **div:** <div> tag is used to write content but it takes space of entire width.

TAGS USED IN TEXT

Tag-Pair	Task
p	Paragraph, sentence
b	To make characters bold
i	Text is shown in italics
em	When you need to emphasize a word but with italics.
strong	When emphasize a word but with bold
sup	Superscript, helpful in formula and footnotes
sub	Subscript, helpful in formula and footnotes
u	Underline a text
small	Smaller text size, like footnote

TAGS IN HTML CONT....

- **img:** `` is used to insert image, src short for source is the path where the file is located.
- **Bullets:** `<ul>` stands for unordered list is used to define bullets. Bullets can be of type circle, square or disc.
- **Numbering:** `<ol>` stands for ordered list is used to define numbering.
- **Links:** `<a>` tag also called anchor tag or hyperlink used to generate link. A link can be text and image. In the link tag we have an attribute – target which define whether the link would open in same or new tab.
- **Table :** `<table>` tag is used to generate table. There are other multiple tags used inside table tag which will discuss in practical task referring the textbook.
- **Video:** `<video>` tag is used to add video.

CASCADING STYLE SHEETS (CSS)

- The CSS is used to add designing in the HTML content. For instances; colors, layout, font style etc.
- In HTML, there are three ways to incorporate CSS; inline, internal, external.
- Inline CSS: in this type, we applying styling on tag in the opening tag.
- Internal CSS: in this type, we apply styling using style tag inside in the head.
- External CSS: in this type, we apply styling by integrating external CSS file link.

JAVASCRIPT (JS)

- JavaScript is a lightweight dynamic programming language that provides interactivity and logic into HTML content.
- Few common examples of JS applications are:
 - *Pop-up boxes, drop-down menus, image carousels, and search bar suggestions.*
 - *Fetching and loading new information in the background without forcing you to manually refresh the page (like a social media feed).*
- JavaScript code can be embedded in HTML with starting and ending tag of `<script>`, in a webpage.
- There is no limitation of where to place the code inside a HTML file.

JAVASCRIPT (JS) CONT....

- **Variable:**

- *A variable is an entity that stores some value for later use.*
- *var, let keywords are used to declare variables in JS.*
- *Variable hold the data of numbers, string, Boolean value.*
- *JS is case sensitive; meaning 'reward' and 'Reward' are two different variables.*
- *Before using any variable, it must be declared otherwise JS will throw error.*

- **Operators:**

- *JS supports arithmetic operators (+, -, x, /).*
- *Modulus (%) operator is an additional operator used to find remainder.*

JAVASCRIPT (JS) CONT....

- **Conditional Statement:**

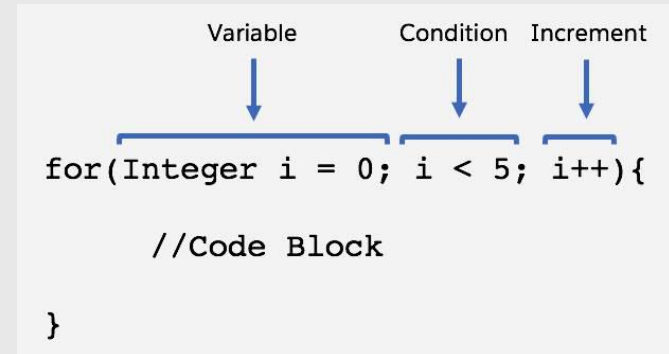
- *Conditional statement is a core feature of programming which help decide a choice based on some criteria.*
- *The keywords 'if', 'else if', 'else' are used to apply conditional statement.*

Operator	Name	Example
==	Is equal	x == y
!=	Is not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal to	x >= y
<=	Less than or equal to	x <= y

JAVASCRIPT (JS) CONT....

- **Iterative Statement:**

- *These are statements which require task to be done in multiple times. For this loops come into action such as for loop.*



The diagram illustrates the components of a for loop. Above the code, three labels are positioned: 'Variable' above 'Integer i = 0', 'Condition' above 'i < 5', and 'Increment' above 'i++'. Blue arrows point from each label to its corresponding part of the loop header. Brackets are used to group these three parts together. The code itself is: `for(Integer i = 0; i < 5; i++){`
`//Code Block`
`}`

- **Nested loops:**

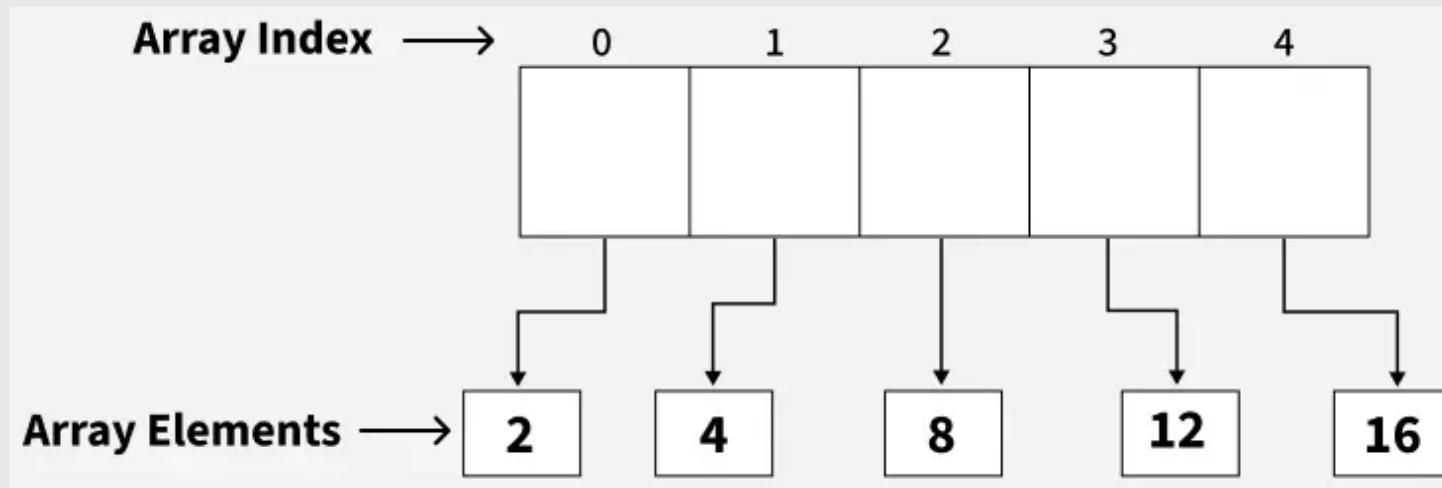
- *A loop inside a loop is called nested loop.*

```
for (let i = 1; i < 4; i++) {  
  for (let k = 1; k < 4; k++) {  
    console.log(i + " , " + k);  
  }  
  console.log("\n");  
}
```

JAVASCRIPT (JS) CONT....

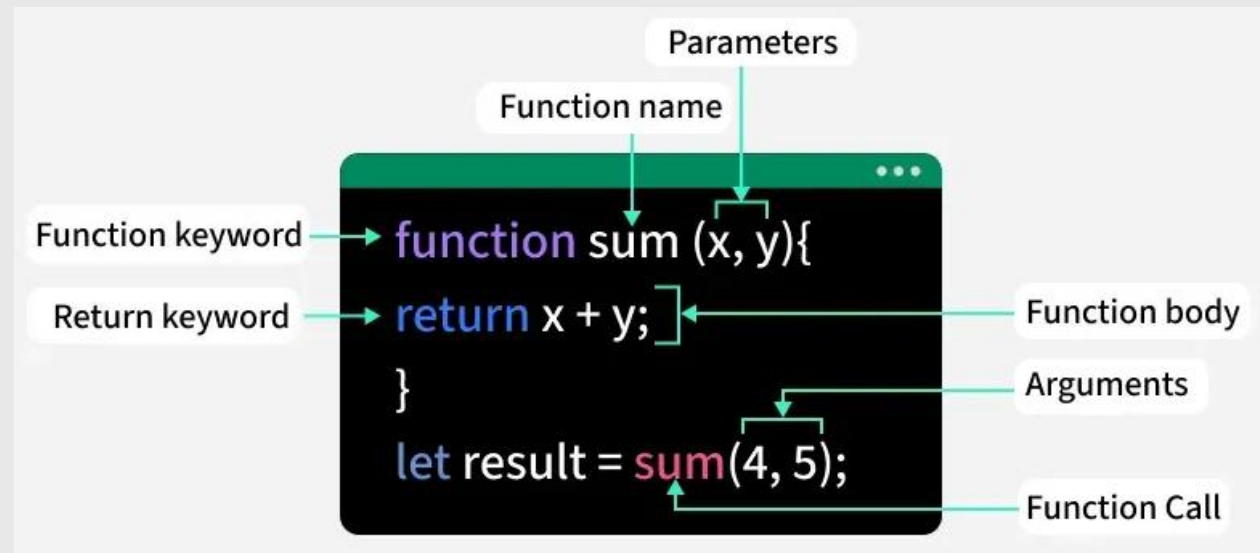
- **Arrays:**

- *An array is a data type that holds a homogeneous set of elements, so you don't need separate variables like num1, num2, ... num25. Instead, declare one array of 25 values and access any value directly by its index. Note that array indexing starts at 0. e.g. `let fruits = ["Apple", "Banana", "Orange"];`*



JAVASCRIPT (JS) CONT....

- **Functions:**
 - A function is a reusable block of code designed to perform a specific task.
 - Languages provide built-in functions like ``prompt()`` and ``input()``.
 - A function has a name, can have parameters and calling.



JAVASCRIPT (JS) CONT....

- **Debug the Code:**
 - *Debugging is a process of finding and fixing errors , or “bugs” in computer program.*

